GOOGLE CLOUD SKILLS | HANDS-ON LAB REPORT

Creating a Persistent Disk

Compute Engine storage · Lab ID: GSP004

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DATE COMPLETED	20 October 2026
PLATFORM	Google Cloud (Compute Engine, Cloud Shell)
LEVEL / DURATION	Introductory / 15 minutes
FINAL SCORE	100 / 100

1. Overview

Compute Engine runs virtual machines on Google infrastructure, and provides **persistent disks** as the primary storage for those VMs. Like a physical hard drive, a persistent disk exists independently of the VM. If the instance is deleted, the disk keeps its data and can be attached to a different instance. There are two types: standard persistent disks and SSD persistent disks.

In this lab I created a VM instance, created a new persistent disk, attached the disk to the running VM, then formatted and mounted it, and set it to mount automatically after a restart.

2. Objectives

- Create a new VM instance and attach a persistent disk
- Find the new disk as a block device inside the VM
- Format the disk and mount it
- Configure /etc/fstab so the disk remounts on restart

3. Resources and Concepts

Resource	Details
Project	qwiklabs-gcp-01-72b6437c59c1
VM instance	gcelab (e2-standard-2, Debian image) with a default 10 GB boot disk
Persistent disk	mydisk – 200 GB, type pd-standard, zone us-east1-d
Device in the VM	/dev/disk/by-id/scsi-0Google_PersistentDisk_persistent-disk-1 (sdb)
Filesystem / mount point	ext4 / /mnt/mydisk

Tool	What it does
gcloud compute disks create	Creates a persistent disk in a zone. It must be in the same zone as the VM it will attach to.
gcloud compute instances attach-disk	Attaches a disk to a running VM. The disk appears as a block device.
mkfs.ext4	Creates an ext4 filesystem. This erases any data on the disk.
mount	Attaches the filesystem to a directory so it can be used.
/etc/fstab	Lists filesystems to mount at boot, so the disk remounts after a restart.

The VM and disk were temporary and were removed when the lab ended.

4. Step-by-Step Walkthrough

Task 1 – Create a VM instance

After setting the region and zone, I created a Compute Engine VM named **gcelab** with only its boot disk:

```
gcloud compute instances create gcelab \
  --zone=$ZONE \
  --machine-type=e2-standard-2 \
  --image-family=<debian image> \
  --image-project=debian-cloud
```

Task 2 – Create a persistent disk

I created a 200 GB disk named **mydisk** in the same zone as the VM (us-east1-d). The output shows type pd-standard and status READY. Cloud Shell also notes that new disks are unformatted, so they must be formatted and mounted before use.

```
gcloud compute disks create mydisk --size=200GB \
  --zone $ZONE
```

Task 3 – Attach, format and mount the disk

I attached the disk to the running VM and connected to the VM over SSH. On the first connection gcloud asked to create an SSH key; I answered **y** and left the passphrase empty.

```
gcloud compute instances attach-disk gcelab --disk mydisk --zone $ZONE
gcloud compute ssh gcelab --zone $ZONE
```

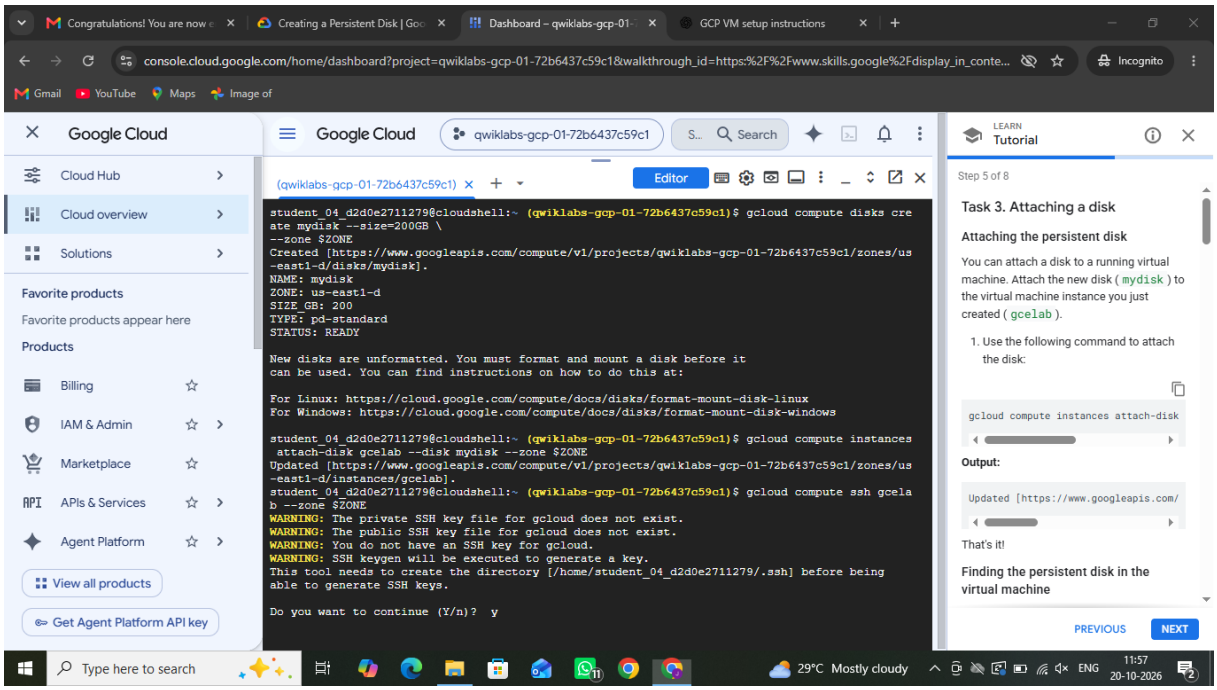


Figure 1: Disk mydisk created (200 GB, pd-standard, READY in us-east1-d), attached to gcelab, and the SSH key creation prompt answered with y.

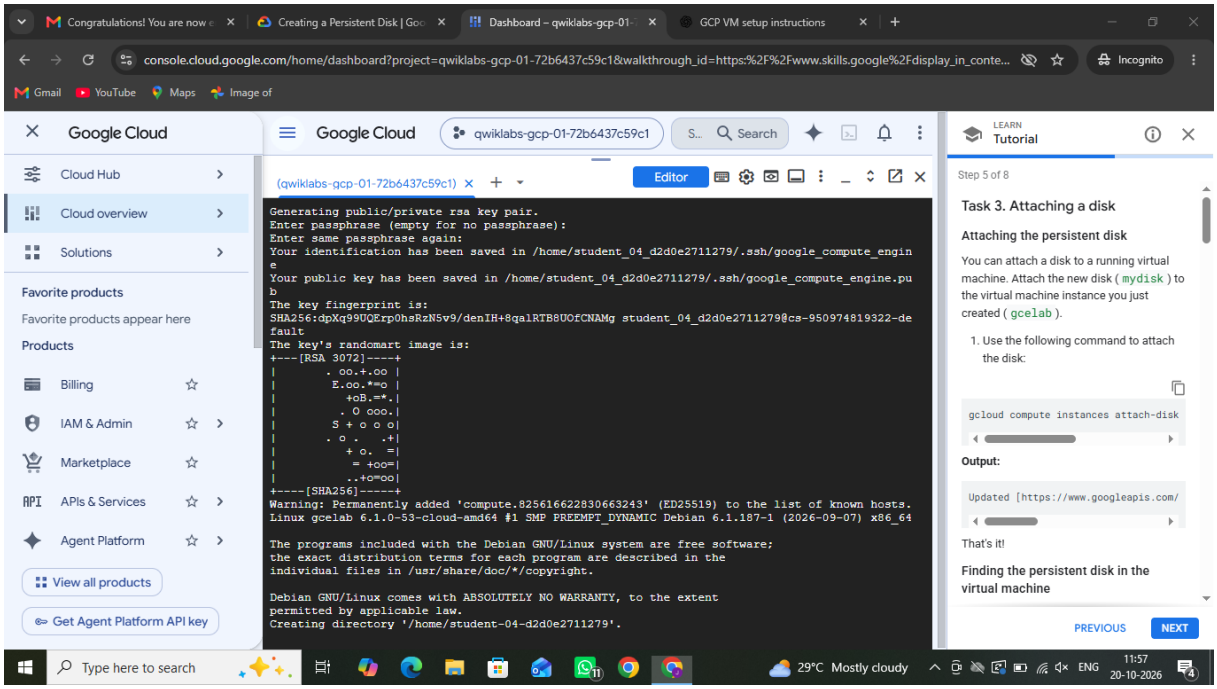


Figure 2: SSH key pair generated and the login to the gcelab VM (Debian GNU/Linux).

Inside the VM, the new disk shows up in `/dev/disk/by-id/` as **scsi-0Google_PersistentDisk_persistent-disk-1**, which points to **sdb**. The boot disk is persistent-disk-0 (sda, with partitions sda1, sda14 and sda15). I created a mount point and formatted the disk with ext4:

```
ls -l /dev/disk/by-id/
sudo mkdir /mnt/mydisk
sudo mkfs.ext4 -F -E lazy_itable_init=0,lazy_journal_init=0,discard \
  /dev/disk/by-id/scsi-0Google_PersistentDisk_persistent-disk-1
```

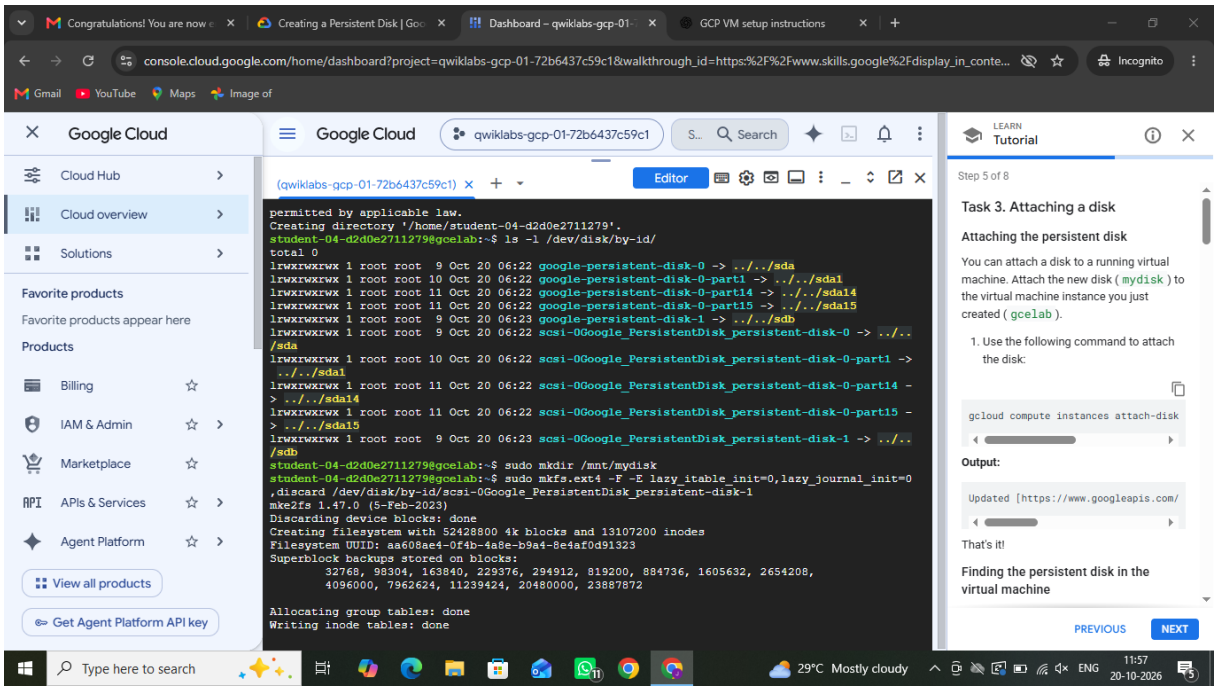


Figure 3: Device listing showing persistent-disk-1 -> sdb, the /mnt/mydisk directory created, and mkfs.ext4 starting (52,428,800 4k blocks, 13,107,200 inodes).

The format finished with the group tables, inode tables, journal and superblocks written. I then mounted the disk with the discard option:

```
sudo mount -o discard,defaults \
  /dev/disk/by-id/scsi-0Google_PersistentDisk_persistent-disk-1 /mnt/mydisk
```

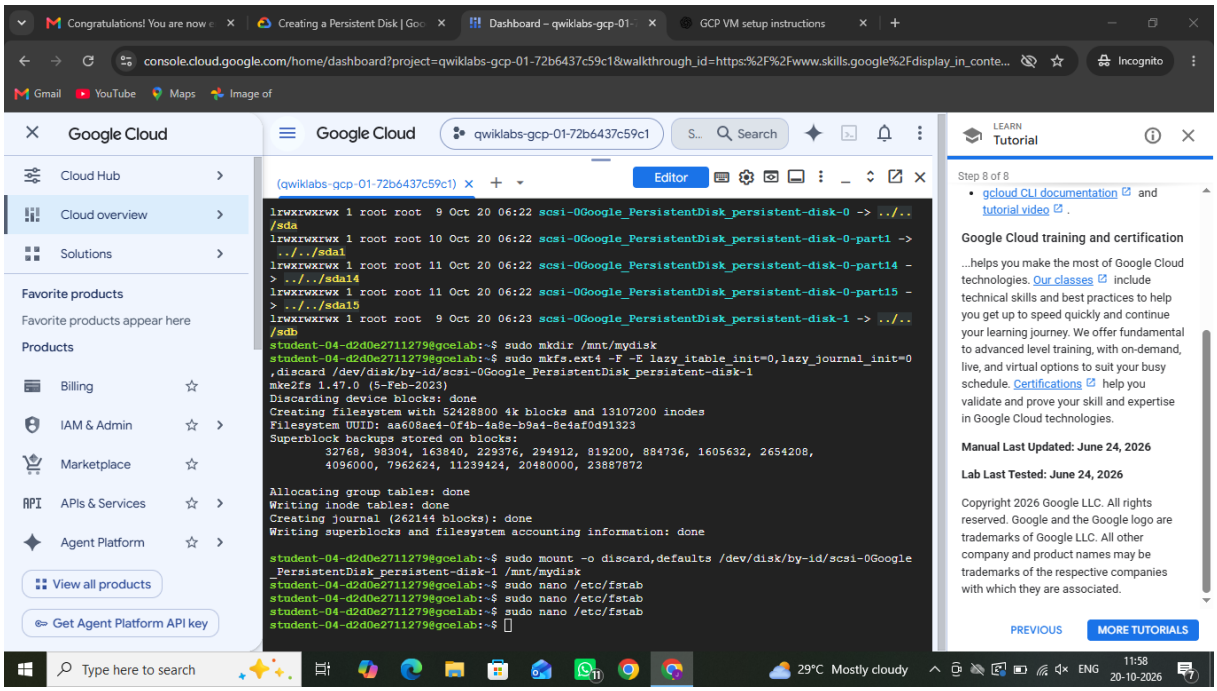


Figure 4: mkfs.ext4 completed (journal of 262,144 blocks created, superblocks written) and the disk mounted at /mnt/mydisk.

Automatic mounting after a restart

By default a disk is not remounted when the VM restarts. I opened `/etc/fstab` in nano and added this line below the PARTUUID entries:

```
/dev/disk/by-id/scsi-0Google_PersistentDisk_persistent-disk-1 /mnt/mydisk ext4 defaults 1 1
```

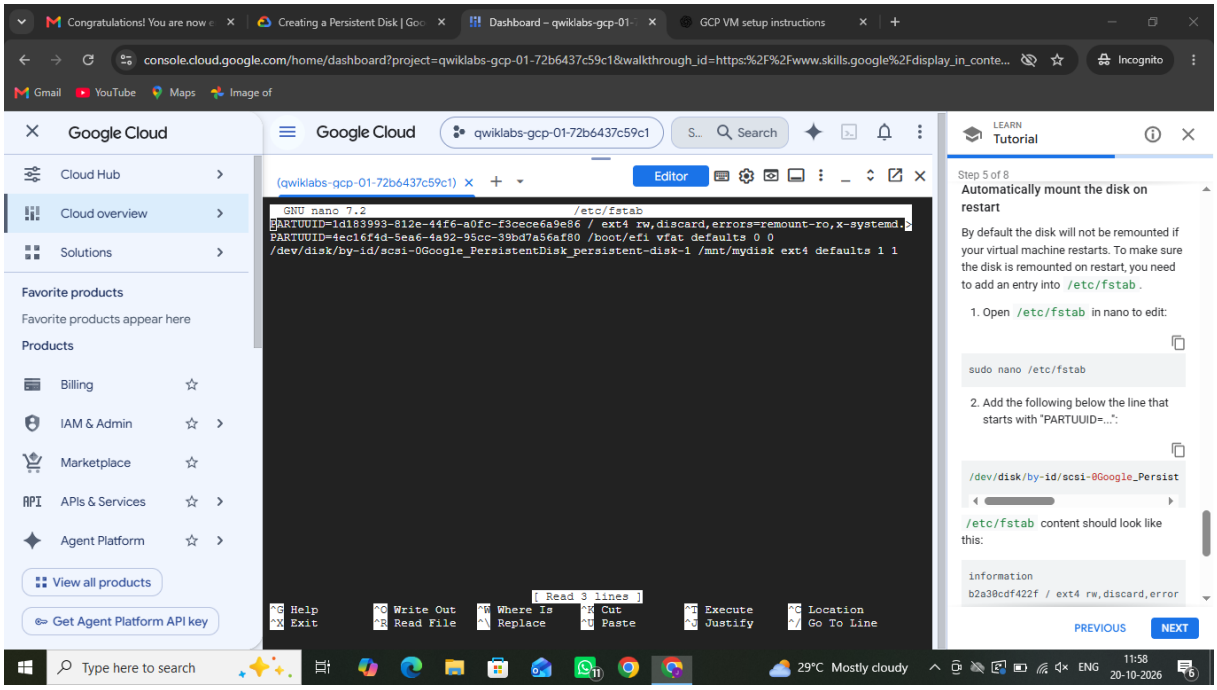


Figure 5: /etc/fstab in nano with the new entry for the persistent disk added after the root and boot/efi PARTUUID lines.

I saved the file with CTRL+O, ENTER and exited with CTRL+X.

5. Lab Completion

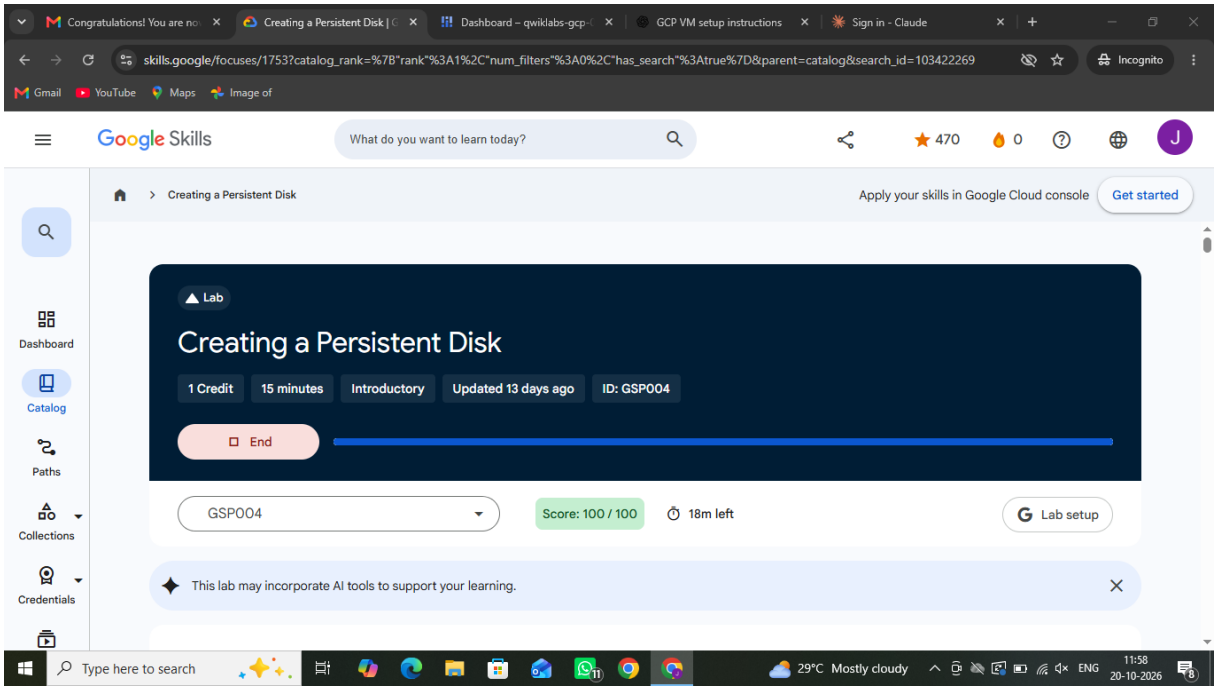


Figure 6: Lab completed with a score of 100/100.

A note on local SSDs

Local SSDs are physically attached to the server that hosts the VM. They give very high IOPS and under 1 ms latency, but they are not automatically replicated, so data can be lost if the host has a problem, and users must back up their data. This lab covered persistent disks only.

6. Key Takeaways

- Persistent disks live independently of the VM. They keep their data if the VM is deleted and can be attached to another instance.
- A disk must be in the **same zone** as the VM it is attached to.
- A new disk is unformatted. It must be formatted (`mkfs.ext4`) and mounted before use, and formatting erases any data on it.
- Inside the VM, an attached disk appears under `/dev/disk/by-id/` with a name such as *scsi-0Google_PersistentDisk_persistent-disk-1*.
- An entry in `/etc/fstab` makes the disk mount automatically after the VM restarts.
- Persistent disks suit durable storage such as database servers. Local SSDs trade durability for performance.

7. Conclusion

The lab was completed with a score of 100/100. I created a VM and a 200 GB persistent disk, attached the disk, formatted it with ext4, mounted it at /mnt/mydisk, and configured /etc/fstab so it remounts on restart.